

Causal Survival Inference from Reconstructed Data: Design-Based and Model-Based Bayesian Nonparametric Approaches

Jonathan Ma Sijia Zhu

April 22, 2026

Abstract

In many clinical settings, individual patient data (IPD) are unavailable, and only reconstructed survival data derived from published Kaplan–Meier curves are accessible. This creates a fundamental challenge for causal inference: treatment effects must be estimated using data that have already undergone an upstream reconstruction process.

We develop a unified pipeline for causal survival analysis using reconstructed IPD, building on outputs from two reconstruction frameworks, KM-GPT and RESOLVE-IPD. Rather than performing reconstruction directly, our focus is on downstream inference. The pipeline integrates three key components: (i) covariate augmentation using trial-level summary statistics, (ii) target-population reweighting to define interpretable causal estimands, and (iii) uncertainty quantification using a design-based Bayesian nonparametric approach based on Dirichlet weighting.

As a model-based extension, we implement a neutral-to-the-right (NTR) hazard-based Bayesian nonparametric survival model, providing an alternative estimate that models the cumulative hazard function directly. This enables a comparison between design-based and model-based nonparametric inference applied to the same reconstructed data.

Simulation studies demonstrate the impact of population heterogeneity on pooled estimates and the stability of the weighting procedure under varying target populations. In a real-data application, we compare KM-GPT and RESOLVE-IPD reconstructions under a common causal framework and show that estimated treatment effects depend on reconstruction method, target population, and survival modeling assumptions. Notably, model-based Bayesian nonparametric estimates are systematically moderated relative to design-based estimates, reflecting smoothing of the survival process through hazard-based modeling and illustrating that different nonparametric formulations can lead to materially different conclusions.

These results emphasize that downstream causal inference from reconstructed survival data is sensitive not only to population definition and uncertainty modeling, but also to upstream reconstruction choices and survival modeling assumptions.

1 Introduction and Motivation

In many clinical and epidemiological studies, access to individual patient data (IPD) is restricted due to privacy, proprietary, or logistical constraints. Instead, published results often provide only aggregate summaries, most commonly in the form of Kaplan–Meier survival curves and limited baseline characteristics. This creates a fundamental challenge for causal inference: treatment effects must be estimated using data that are not directly observed, but instead reconstructed from incomplete information.

Recent advances have made it possible to approximate IPD from published survival curves with increasing accuracy. Methods such as KM-GPT [1] and RESOLVE-IPD [2] provide automated pipelines for extracting individual-level survival times from Kaplan–Meier plots, enabling downstream statistical analysis even when raw data are unavailable. These tools have been applied to oncology trials such as ATTRACTION-3 [3], ESCORT [4], and KEYNOTE-181 [5], where survival outcomes are often reported only through aggregated visualizations.

Despite these advances, reconstructed IPD introduces new challenges for causal inference. First, different reconstruction methods may yield systematically different datasets, leading to variation in downstream estimates even when applied to the same study. Second, reconstructed survival data typically lack individual-level covariates, as baseline characteristics are reported only at the aggregate level. Third, when combining multiple studies, differences in patient populations lead to heterogeneous mixtures that do not correspond to a well-defined causal estimand. These issues are closely related to broader challenges in combining aggregate and individual-level data in a causally interpretable way [6], as well as to inference under covariate shift using summary statistics [7].

Addressing these challenges requires a framework that goes beyond reconstruction and explicitly accounts for downstream inference. In particular, it is necessary to (i) incorporate auxiliary summary statistics to recover plausible covariate structures, (ii) define a target population for causal interpretation, and (iii) quantify uncertainty arising from both population heterogeneity and incomplete information.

In this work, we focus on causal survival inference using reconstructed IPD as input, rather than on the reconstruction process itself. Building on outputs from KM-GPT and RESOLVE-IPD, we develop a unified pipeline that integrates covariate augmentation, target-population reweighting, and Bayesian nonparametric uncertainty quantification. The goal is to estimate treatment-specific survival probabilities at a fixed time horizon and their contrast under a well-defined population.

To capture uncertainty in the target population, we adopt a design-based Bayesian nonparametric approach using Dirichlet weighting, which places a flexible prior over the empirical distribution of reconstructed individuals. As a complementary extension, we also consider a model-based Bayesian nonparametric approach based on a neutral-to-the-right (NTR) hazard formulation, which directly models the cumulative hazard function. This allows us to compare two distinct nonparametric paradigms: uncertainty over weights versus uncertainty over the survival process.

We evaluate the proposed framework through both simulation and real-data analyses based on reconstructed data from ATTRACTION-3, ESCORT, and KEYNOTE-181. By comparing KM-GPT and RESOLVE-IPD under a common downstream pipeline, we iso-

late the impact of reconstruction, and by incorporating both design-based and model-based Bayesian nonparametric methods, we assess the sensitivity of conclusions to different formulations of uncertainty.

Overall, this work emphasizes that causal inference from reconstructed survival data is inherently a multi-stage problem. Reconstruction, population definition, and modeling choices all contribute to the final estimate, and must be considered jointly to produce reliable and interpretable conclusions.

2 Related Work

TODO

3 Methodology

3.1 Problem Setup and Causal Estimand

We consider a survival analysis setting in which only reconstructed individual patient data (IPD) are available, obtained from published Kaplan–Meier curves using external tools such as KM-GPT and RESOLVE-IPD. Let T denote the event time (e.g., time to death), $A \in \{0, 1\}$ the treatment indicator, and X a vector of baseline covariates.

We adopt the potential outcomes framework for causal inference. For each individual, let $T^{(1)}$ and $T^{(0)}$ denote the potential event times under treatment and control, respectively. The corresponding survival functions are defined as

$$S_a(t) = \mathbb{P}(T^{(a)} > t), \quad a \in \{0, 1\}.$$

Our primary estimand is the causal survival contrast at a fixed time horizon t_0 :

$$\Delta(t_0) = S_1(t_0) - S_0(t_0).$$

This estimand represents the difference in survival probabilities between treatment arms at time t_0 under a specified target population. In randomized trials, treatment assignment is independent of potential outcomes conditional on baseline covariates, allowing identification of $S_a(t)$ within each study. However, when reconstructed IPD from multiple studies are pooled, the resulting dataset reflects a mixture of underlying populations with differing covariate distributions.

A naive pooled estimator computes $\Delta(t_0)$ by aggregating all reconstructed individuals and applying a Kaplan–Meier estimator within treatment groups. This approach implicitly targets a mixture population and may not correspond to a clinically meaningful estimand. In particular, differences in baseline covariates across studies induce covariate shift, making the naive estimate difficult to interpret causally.

To address this, we define a target population characterized by a distribution F_X^* over baseline covariates. The goal is to estimate

$$S_a^*(t_0) = \mathbb{E}_{X \sim F_X^*} [\mathbb{P}(T > t_0 \mid A = a, X)],$$

and the corresponding contrast

$$\Delta^*(t_0) = S_1^*(t_0) - S_0^*(t_0).$$

In practice, F_X^* is specified through auxiliary summary statistics reported in trial publications, such as mean age, sex proportions, or biomarker prevalence. Since individual-level covariates are not directly observed, they must be approximated through a covariate augmentation procedure that matches these summaries. The resulting pseudo-individual data are then reweighted to align with the target population, enabling estimation of the desired causal survival contrast.

This formulation highlights three key challenges: (i) reconstructed survival data are only approximations of the true IPD, (ii) covariate information is incomplete and must be inferred from summaries, and (iii) causal estimands depend on the choice of target population. The methodology developed in this work addresses these challenges through a combination of augmentation, reweighting, and Bayesian nonparametric inference.

3.2 Reconstruction of Individual Patient Data

In this work, individual patient data (IPD) are not directly observed, but instead obtained from existing reconstruction pipelines applied to published Kaplan–Meier curves. Specifically, we use reconstructed datasets produced by KM-GPT [1] and RESOLVE-IPD [2], which provide approximate individual-level survival times and censoring indicators derived from aggregate survival plots.

These reconstruction methods aim to recover pseudo-IPD that are consistent with the reported Kaplan–Meier curves and, when available, additional information such as numbers at risk or subgroup breakdowns. The resulting datasets consist of tuples (T_i, δ_i, A_i) for each reconstructed individual i , where T_i denotes the observed time, δ_i the event indicator, and A_i the treatment assignment.

While reconstruction accuracy has been shown to be high under ideal conditions, it is important to emphasize that reconstructed IPD are inherently approximate. Different reconstruction methods may produce systematically different datasets even when applied to the same underlying study, due to differences in interpolation, censoring assumptions, and use of auxiliary information. As a result, downstream analyses based on reconstructed data may vary depending on the chosen reconstruction pipeline.

In this project, we treat reconstructed IPD as the input to the causal inference pipeline and focus on downstream statistical analysis. By applying a common augmentation, weighting, and inference framework to both KM-GPT and RESOLVE-IPD outputs, we are able to isolate the impact of reconstruction on estimated treatment effects. This allows us to assess the sensitivity of causal conclusions to upstream reconstruction choices, without directly modifying or re-implementing the reconstruction procedures themselves.

3.3 Covariate Augmentation from Auxiliary Summaries

Reconstructed individual patient data (IPD) obtained from Kaplan–Meier curves typically include only survival times, censoring indicators, and treatment assignments. However,

causal inference in heterogeneous populations requires adjustment for baseline covariates, which are generally not available at the individual level. Instead, trial publications report only aggregate summaries, such as mean age, proportions of male patients, or prevalence of clinical characteristics.

To incorporate this information, we construct synthetic individual-level covariates that are consistent with the reported auxiliary summaries. Let X denote a vector of baseline covariates, including both continuous and binary variables. For each trial (and subgroup, when available), we generate pseudo-covariates $\{X_i\}_{i=1}^n$ such that their empirical moments approximately match the reported summaries.

For continuous variables (e.g., age), values are generated from parametric distributions calibrated to match reported means. For binary variables (e.g., sex, biomarker status), values are sampled from Bernoulli distributions with probabilities equal to the reported proportions. When multiple covariates are present, they are generated independently conditional on the reported marginal summaries.

This procedure yields an augmented dataset of the form $(T_i, \delta_i, A_i, X_i)$ for each reconstructed individual. While the marginal distributions of the simulated covariates are aligned with published summaries, the joint dependence structure between covariates is not recovered, as it is not identifiable from aggregate information alone.

To assess the validity of the augmentation procedure, we compare the empirical moments of the simulated covariates to the target summaries across trials and subgroups. In practice, the augmented covariates closely match the reported values, with small discrepancies attributable to finite-sample variability.

This covariate augmentation step enables downstream adjustment for population differences across studies and provides the necessary inputs for target-population reweighting. However, it also introduces an additional layer of uncertainty, as the true individual-level covariate structure is not observed. This limitation is addressed in subsequent stages through weighting and Bayesian nonparametric inference.

3.4 Target Population and Reweighting

When reconstructed IPD from multiple trials are pooled, the resulting dataset represents a mixture of underlying study populations with different baseline covariate distributions. As a result, naive pooled estimates do not correspond to a well-defined causal estimand, but instead reflect an implicit average over heterogeneous populations.

To obtain interpretable causal estimates, we define a target population characterized by a distribution F_X^* over baseline covariates. In practice, F_X^* is specified through a set of desired summary statistics, such as mean age or proportions of key clinical characteristics. These target values may be motivated by clinical relevance or by external population benchmarks.

Let $\{X_i\}_{i=1}^n$ denote the augmented covariates for reconstructed individuals. We seek a set of nonnegative weights $\{w_i\}_{i=1}^n$ such that the weighted empirical moments match the target summaries. Formally, the weights are chosen to satisfy constraints of the form

$$\sum_{i=1}^n w_i X_i \approx \mu^*, \quad \sum_{i=1}^n w_i = 1,$$

where μ^* represents the target moments.

In this work, we implement reweighting via a moment-matching approach, which adjusts the contribution of each individual so that the weighted sample approximates the target population. The resulting weights define a new empirical distribution over the reconstructed individuals, enabling estimation of treatment-specific survival functions under the target population.

Given weights $\{w_i\}$, the causal survival probabilities at time t_0 are estimated as

$$S_a^*(t_0) = \sum_{i:A_i=a} w_i \mathbf{1}(T_i > t_0),$$

with contrast $\Delta^*(t_0) = S_1^*(t_0) - S_0^*(t_0)$.

An important diagnostic for the stability of the weighting procedure is the effective sample size (ESS), defined as

$$\text{ESS} = \frac{1}{\sum_{i=1}^n w_i^2}.$$

Small ESS values indicate that the weighted estimate is dominated by a small subset of individuals, which may lead to high variance and reduced reliability.

This reweighting step allows us to decouple the causal estimand from the observed mixture of study populations and instead target a clinically meaningful population. However, it also introduces sensitivity to the choice of target distribution and the feasibility of the moment-matching constraints, which is further explored in the simulation and empirical analyses.

3.5 Design-Based Bayesian Nonparametric Inference

The reweighting procedure described above produces a point estimate of the causal survival contrast under the target population. However, this estimate does not account for uncertainty arising from finite sample size, covariate augmentation, and the implicit variability in the weighting scheme. To quantify this uncertainty, we adopt a design-based Bayesian nonparametric approach.

Specifically, we place a Dirichlet prior over the weights assigned to reconstructed individuals. Let $w = (w_1, \dots, w_n)$ denote the weights on the n augmented individuals. We assume

$$w \sim \text{Dirichlet}(\alpha_1, \dots, \alpha_n),$$

where typically $\alpha_i = \alpha$ for all i , corresponding to a symmetric prior. This formulation is closely related to the Bayesian bootstrap, which treats the empirical distribution as random and induces uncertainty over reweighting schemes.

To incorporate the target population constraints, we condition on weights that approximately satisfy the moment-matching conditions. In practice, this is implemented by repeatedly sampling candidate weight vectors from the Dirichlet prior and retaining those that produce weighted covariate summaries close to the target values, or by applying a projection step that adjusts sampled weights to satisfy the constraints.

Given a sampled weight vector $w^{(s)}$, the treatment-specific survival probabilities at time t_0 are computed as

$$S_a^{(s)}(t_0) = \sum_{i:A_i=a} w_i^{(s)} \mathbf{1}(T_i > t_0),$$

and the corresponding causal contrast is

$$\Delta^{(s)}(t_0) = S_1^{(s)}(t_0) - S_0^{(s)}(t_0).$$

Repeating this procedure for $s = 1, \dots, S$ yields a collection of posterior draws $\{\Delta^{(s)}(t_0)\}_{s=1}^S$, which approximate the posterior distribution of the treatment effect under the target population. From these samples, we compute posterior summaries such as the mean, standard deviation, and credible intervals.

This design-based Bayesian nonparametric approach has two key advantages. First, it avoids specifying a parametric model for the survival distribution, relying instead on the empirical support of the reconstructed data. Second, it directly captures uncertainty in the weighting scheme, which is a primary source of variability when targeting external populations.

However, this approach does not model the survival process itself and therefore treats observed survival times as fixed. As a result, it captures uncertainty in population representation but not in the underlying survival distribution. This motivates the complementary model-based Bayesian nonparametric approach introduced in the next subsection.

3.6 Model-Based Bayesian Nonparametric Survival Modeling (NTR)

While the design-based Bayesian nonparametric approach captures uncertainty through reweighting of observed data, it does not model the survival distribution itself. To assess sensitivity to modeling assumptions, we consider a complementary model-based Bayesian nonparametric approach based on neutral-to-the-right (NTR) processes.

Under an NTR formulation, the cumulative hazard function is modeled as a stochastic process with independent increments. In practice, we approximate this process using a piecewise-constant hazard over a grid of time intervals. For treatment arm $a \in \{0, 1\}$ and interval j , the hazard increment is modeled as

$$\lambda_{a,j} \sim \text{Gamma}(\alpha, \beta),$$

with posterior updates determined by observed event counts and total exposure within each interval.

The survival function is then obtained as

$$S_a(t) = \exp \left(- \sum_{j:t_j \leq t} \lambda_{a,j} \Delta t_j \right),$$

yielding a flexible, nonparametric representation of the survival process.

Posterior inference is performed by sampling hazard increments independently across intervals and constructing survival curves for each draw. The causal contrast is computed as

$$\Delta^{(s)}(t_0) = S_1^{(s)}(t_0) - S_0^{(s)}(t_0).$$

This approach differs from the design-based method in that uncertainty propagates through the survival model itself, rather than through weights on observed outcomes. The

NTR formulation provides a direct representation of the survival process via the cumulative hazard and induces smoothing through the piecewise hazard structure.

As a result, differences between design-based and model-based estimates reflect both statistical uncertainty and sensitivity to assumptions about the underlying survival dynamics.

4 Simulation Study

We evaluate the proposed framework in a controlled simulation setting designed to mirror the key challenges of causal survival inference using reconstructed data. The simulation study serves three primary purposes: (i) to validate the accuracy of survival reconstruction under ideal conditions, (ii) to assess the impact of heterogeneity across studies when pooling data, and (iii) to examine the stability of target-population reweighting under varying degrees of covariate shift.

The simulation framework generates synthetic individual patient data with known survival mechanisms, treatment assignments, and baseline covariates. Kaplan–Meier curves are constructed from the true data and treated as the only observable inputs, mimicking real-world settings in which raw IPD are unavailable. Reconstructed datasets are then obtained using the same pipeline applied in the empirical analysis, allowing direct comparison between true and reconstructed quantities.

We consider both single-trial and multi-trial settings. In the single-trial case, reconstruction accuracy is evaluated by comparing survival curves and treatment effects computed from true and reconstructed data. In the multi-trial setting, heterogeneity is introduced through differences in covariate distributions and survival dynamics across studies. Reconstructed datasets are pooled to examine how naive aggregation, target-population reweighting, and Bayesian nonparametric inference behave under population mixing.

Finally, we conduct stress tests by specifying increasingly challenging target populations that differ from the observed mixture of studies. These experiments allow us to evaluate the stability of the weighting procedure, using metrics such as effective sample size (ESS) and posterior variability, and to understand the limits of the proposed approach under extreme covariate shift.

Together, these simulation experiments provide a systematic validation of the pipeline and offer insight into how reconstruction, heterogeneity, and weighting interact to influence causal survival estimates.

4.1 Single-Trial Reconstruction Validation

We begin by evaluating the accuracy of the reconstruction step in a controlled single-trial setting. The goal of this experiment is to assess whether survival quantities computed from reconstructed data closely match those obtained from the true underlying individual patient data (IPD).

We generate synthetic survival data for a single randomized trial, including event times T , censoring indicators δ , and treatment assignments $A \in \{0, 1\}$. Kaplan–Meier curves are constructed from the true data and treated as the only observable inputs, mimicking the

information typically available in published studies. Reconstructed IPD are then obtained using the same pipeline applied in the empirical analysis.

Reconstruction accuracy is assessed by comparing survival curves and treatment-specific survival probabilities between the true and reconstructed datasets. In particular, we evaluate the root mean squared error (RMSE) between Kaplan–Meier estimates and compare survival probabilities at a fixed time horizon t_0 . Because the true data-generating process is known, this setting provides a direct benchmark for assessing reconstruction fidelity.

The results show that the reconstruction procedure achieves near-perfect recovery of the survival curve, with negligible RMSE and minimal differences in survival probabilities at t_0 . In addition, the number of reconstructed individuals closely matches the true sample size, indicating that the overall structure of the data is preserved. Representative simulation-based posterior behavior is shown in (Fig. A2).

These findings confirm that, under ideal conditions with clean inputs, reconstruction does not introduce meaningful distortion into survival quantities. This validates the use of reconstructed IPD as input for downstream causal analysis and allows subsequent experiments to focus on challenges arising from heterogeneity, population definition, and uncertainty quantification rather than reconstruction error.

4.2 Multi-Trial Heterogeneity and Pooling

We extend the analysis to a multi-trial setting to study the impact of heterogeneity across studies when reconstructed datasets are pooled. This setting reflects real-world meta-analytic scenarios, where trials differ in baseline covariate distributions, sample sizes, and underlying survival dynamics.

We simulate multiple trials, each with its own covariate distribution and treatment effect. For each trial, individual patient data are generated and corresponding Kaplan–Meier curves are constructed. As in the single-trial setting, only the Kaplan–Meier curves are treated as observed, and reconstructed IPD are obtained using the same reconstruction pipeline.

The reconstructed datasets are then pooled across trials to form a combined dataset. Treatment-specific survival probabilities are estimated using standard Kaplan–Meier estimators within treatment groups, yielding a naive pooled estimate of the causal contrast $\Delta(t_0)$.

The results show substantial variation in treatment effects across trials, driven by differences in underlying populations and data-generating mechanisms. When pooled, these heterogeneous effects are averaged together, producing a combined estimate that does not correspond to any single study population. Instead, the naive estimator implicitly targets a mixture population defined by the relative sizes and covariate distributions of the contributing trials.

This experiment highlights a key limitation of naive pooling: without adjustment, the resulting estimate lacks a clear causal interpretation. The presence of heterogeneity across studies motivates the need for target-population reweighting, which allows treatment effects to be defined with respect to a specified covariate distribution. It also underscores that differences in population composition can be as influential as differences in reconstruction, reinforcing the importance of explicitly accounting for both sources of variability in downstream inference.

4.3 Stress Testing Target Populations

To evaluate the robustness of the reweighting procedure, we conduct stress tests under increasingly challenging target populations. The goal is to assess how sensitive the estimated treatment effects are to the choice of target covariate distribution and to identify conditions under which the weighting approach becomes unstable.

Starting from the pooled reconstructed dataset, we define a sequence of target populations characterized by covariate summaries that progressively deviate from those observed in the data. These targets are specified through shifts in means and proportions of key covariates, such as age, biomarker prevalence, and disease severity. Each target population therefore represents a different extrapolation scenario, ranging from well-supported regions of the covariate space to more extreme configurations.

For each target population, we compute weights that match the specified summary statistics and evaluate the resulting causal survival contrast $\Delta^*(t_0)$. In addition to point estimates, we monitor the effective sample size (ESS) and the variability of the posterior distribution obtained through the design-based Bayesian nonparametric procedure. Posterior changes across target populations are shown in (Fig. A3), while the corresponding decline in effective sample size is shown in (Fig. A4).

The results reveal a clear trade-off between representativeness and stability. When the target population is close to the observed mixture of trials, the weights are relatively uniform, the ESS remains high, and posterior uncertainty is limited. As the target population moves further from the observed data, the weights become increasingly concentrated on a small subset of individuals, leading to a reduction in ESS and a corresponding increase in uncertainty.

In extreme cases, the weighting procedure effectively extrapolates beyond the support of the data, resulting in highly variable estimates that are sensitive to a small number of observations. These findings highlight the importance of choosing target populations that are well-supported by the available data and demonstrate that ESS provides a useful diagnostic for assessing the reliability of weighted causal estimates.

Overall, the stress tests illustrate the limits of reweighting-based inference and emphasize that uncertainty arising from population mismatch must be carefully accounted for in practice.

5 Real Data Application

We apply the proposed framework to reconstructed individual patient data (IPD) from three oncology trials: ATTRACTION-3 [3], ESCORT [4], and KEYNOTE-181 [5]. For each study, survival data are obtained from existing reconstruction pipelines, specifically KM-GPT [1] and RESOLVE-IPD [2]. These datasets provide event times, censoring indicators, and treatment assignments, but do not include individual-level covariates.

The objective of this application is to evaluate how downstream causal inference is affected by reconstruction method, target population definition, and modeling assumptions. By applying a common augmentation, reweighting, and inference pipeline to both KM-GPT and RESOLVE-IPD outputs, we isolate the impact of reconstruction while holding all other

components fixed.

We first augment the reconstructed datasets with synthetic covariates based on trial-level summary statistics, ensuring consistency with reported baseline characteristics. We then define a target population and compute weights that align the pooled dataset with the desired covariate distribution. Using these weights, we estimate treatment-specific survival probabilities and their contrast at a fixed time horizon.

Uncertainty is quantified using a design-based Bayesian nonparametric approach based on Dirichlet weighting. In addition, for the KM-GPT pipeline, we implement a model-based Bayesian nonparametric survival model based on a neutral-to-the-right (NTR) hazard formulation to assess sensitivity to assumptions about the underlying survival distribution.

This real-data application allows us to compare reconstruction pipelines, evaluate the impact of target-population adjustment, and examine differences between design-based and model-based Bayesian nonparametric inference in a realistic setting.

5.1 KM-GPT Reconstruction Pipeline

We begin by applying the downstream inference pipeline to reconstructed individual patient data (IPD) obtained from KM-GPT [1]. The KM-GPT output provides pseudo-individual records consisting of event times T_i , censoring indicators δ_i , and treatment assignments A_i for each study. These reconstructed datasets serve as the starting point for all subsequent analysis. The reconstructed survival patterns across trials and subgroups are shown in (Fig. A1).

Because individual-level covariates are not included in the reconstruction output, we augment the data using trial-level summary statistics. Synthetic covariates are generated for each individual to match reported baseline characteristics, resulting in an augmented dataset $(T_i, \delta_i, A_i, X_i)$. This enables adjustment for population heterogeneity across studies.

We then pool the augmented datasets across trials and compute a naive estimate of the causal survival contrast $\Delta(t_0)$ using Kaplan–Meier estimators within treatment groups. As expected, this estimate reflects a mixture of underlying study populations and does not correspond to a well-defined target population.

To address this, we define a target population through specified covariate summaries and compute weights that align the pooled dataset with this target. The weighted estimator produces a treatment effect that is interpretable with respect to the chosen population, and in our analysis, this adjustment leads to a substantial change in the estimated survival contrast relative to the naive estimate.

Uncertainty is quantified using the design-based Bayesian nonparametric approach described earlier, yielding posterior summaries for $\Delta(t_0)$ under the target population. In addition, we apply a model-based Bayesian nonparametric survival model based on a neutral-to-the-right (NTR) hazard formulation to the KM-GPT data, providing an alternative estimate that directly models the survival process.

Overall, the KM-GPT pipeline serves as the primary analysis in this study, illustrating how reconstruction, covariate augmentation, reweighting, and Bayesian nonparametric inference combine to produce causal survival estimates from reconstructed data.

5.2 RESOLVE-IPD Reconstruction Pipeline

We next apply the same downstream inference pipeline to reconstructed individual patient data (IPD) obtained from RESOLVE-IPD [2]. As with KM-GPT, the RESOLVE-IPD output provides pseudo-individual records consisting of event times T_i , censoring indicators δ_i , and treatment assignments A_i for each study.

To ensure a fair comparison, all subsequent steps are held fixed relative to the KM-GPT analysis. We augment the reconstructed data with synthetic covariates generated from trial-level summary statistics, producing an augmented dataset $(T_i, \delta_i, A_i, X_i)$ that is consistent with reported baseline characteristics. The augmented datasets are then pooled across trials.

A naive estimate of the causal survival contrast $\Delta(t_0)$ is computed using Kaplan–Meier estimators within treatment groups. As in the KM-GPT pipeline, this estimate reflects a mixture of study populations and lacks a clear causal interpretation.

We then define the same target population used in the KM-GPT analysis and compute weights that align the pooled RESOLVE-IPD dataset with this target. The resulting weighted estimator provides a treatment effect that is directly comparable across reconstruction methods. In our analysis, the weighted estimate differs substantially from both the naive estimate and the corresponding KM-GPT result, highlighting sensitivity to the reconstruction procedure.

Uncertainty is quantified using the same design-based Bayesian nonparametric approach based on Dirichlet weighting, yielding posterior summaries for $\Delta(t_0)$ under the target population. Unlike the KM-GPT pipeline, we do not apply the model-based NTR extension to the RESOLVE-IPD data, allowing us to isolate the effect of reconstruction separately from survival modeling assumptions.

This parallel analysis enables a direct comparison between KM-GPT and RESOLVE-IPD under a common causal framework, isolating the impact of reconstruction on downstream inference.

5.3 Covariate Augmentation and Validation

Because reconstructed individual patient data (IPD) do not include baseline covariates, we augment both KM-GPT and RESOLVE-IPD datasets using trial-level summary statistics. For each study (and subgroup, when available), we generate synthetic covariates for reconstructed individuals so that their empirical moments match reported summaries, including quantities such as mean age and proportions of key clinical characteristics.

The augmentation procedure is applied consistently across reconstruction pipelines to ensure comparability. For continuous variables, values are generated to match reported means, while for binary variables, values are sampled to match reported proportions. This produces augmented datasets of the form $(T_i, \delta_i, A_i, X_i)$ for each reconstructed individual.

To assess the validity of this procedure, we compare the empirical moments of the simulated covariates with the corresponding target summaries across trials and subgroups. In particular, we examine whether the realized covariate means and proportions closely align with the reported values used to generate them. A representative validation plot comparing target and realized covariate summaries is shown in (Fig. A5).

The results show that the augmented covariates closely match the target summaries, with

deviations attributable to finite-sample variability. Diagnostic plots comparing target and realized values indicate that most quantities lie near the identity line, suggesting accurate reproduction of the intended marginal distributions.

This validation step supports the use of augmented covariates for downstream analysis. While the joint dependence structure between covariates is not recovered, the marginal alignment is sufficient for implementing target-population reweighting and enables meaningful comparison of causal estimates across reconstruction pipelines.

6 Results

6.1 Comparison of Reconstruction Pipelines

We compare results obtained from the KM-GPT and RESOLVE-IPD reconstruction pipelines under a common downstream causal inference framework. In both cases, the same covariate augmentation procedure, target population, weighting method, and design-based Bayesian nonparametric inference are applied, allowing us to isolate the impact of the reconstruction method.

Naive pooled estimates differ substantially across pipelines, with KM-GPT producing a larger estimated survival contrast $\Delta(t_0)$ than RESOLVE-IPD. This confirms that reconstruction alone introduces variability into downstream inference, even when applied to the same underlying studies.

Target-population weighting further amplifies these differences. Under the KM-GPT pipeline, weighting increases the estimated treatment effect, while under the RESOLVE-IPD pipeline, it decreases it. This demonstrates that treatment effects are highly sensitive to both the reconstruction method and the population over which they are defined.

Bayesian nonparametric uncertainty quantification highlights additional differences. The KM-GPT-based posterior suggests a consistently positive treatment effect, with credible intervals entirely above zero. In contrast, the RESOLVE-IPD-based posterior is wider and includes zero, indicating greater uncertainty and weaker evidence for a treatment benefit.

Method	Naive	Weighted	BNP Mean	BNP 2.5%	BNP 97.5%	NTR Mean
KM-GPT	0.169	0.234	0.205	0.052	0.361	0.157
RESOLVE-IPD	0.125	0.093	0.121	-0.017	0.252	—

Table 1: Comparison of treatment effect estimates $\Delta(t_0)$ across reconstruction pipelines and inference methods. Naive estimates correspond to pooled Kaplan–Meier differences, while weighted estimates adjust for the target population. BNP results report posterior means and 95% credible intervals from the design-based Bayesian nonparametric approach. The NTR column provides model-based Bayesian nonparametric estimates for the KM-GPT pipeline.

For the KM-GPT data, we additionally apply a model-based Bayesian nonparametric survival model based on a neutral-to-the-right (NTR) hazard formulation. The resulting estimate is systematically moderated relative to both the weighted and design-based BNP estimates, indicating that conclusions are sensitive to assumptions about the survival process.

Notably, the NTR estimate is closer in magnitude to the RESOLVE-IPD results, suggesting that model-based inference may partially reconcile differences observed across reconstruction pipelines. A summary comparison across reconstruction pipelines and inference methods is shown in (Fig. A6).

Overall, these results demonstrate that reconstruction method, population definition, and modeling assumptions all contribute meaningfully to the final causal estimate.

6.2 Validation of Covariate Augmentation

We evaluate the accuracy of the covariate augmentation procedure by comparing target auxiliary summaries to the realized covariate distributions in the augmented datasets.

For each trial and subgroup, we compute empirical means and proportions of the simulated covariates and compare them to the reported summary statistics used as targets. Across all covariates, the realized values closely match the target summaries, with only minor deviations due to sampling variability.

Diagnostic plots comparing target and realized values show that most points lie near the identity line, indicating strong agreement between intended and simulated distributions. This confirms that the augmentation procedure successfully reproduces the marginal covariate structure reported in the original studies.

These results support the validity of the covariate augmentation step. Although the joint dependence structure between covariates is not recovered, the accurate matching of marginal distributions is sufficient for implementing target-population reweighting and enables meaningful downstream causal inference.

6.3 Impact of Target-Population Weighting

Target-population weighting has a substantial impact on the estimated treatment effect. When applied to the pooled reconstructed data, weighting shifts the empirical distribution of covariates to align with a specified target population, thereby redefining the estimand.

Under the KM-GPT reconstruction, weighting increases the estimated survival contrast $\Delta(t_0)$ relative to the naive pooled estimate, indicating that the target population places greater weight on individuals with higher treatment benefit. In contrast, under the RESOLVE-IPD reconstruction, weighting decreases the estimated effect, suggesting that the target population emphasizes individuals with lower treatment benefit.

These opposing shifts highlight that the direction and magnitude of the treatment effect are sensitive to both the underlying reconstruction and the chosen target population. In particular, the naive pooled estimate implicitly targets a mixture population determined by the composition of the contributing studies, whereas the weighted estimate corresponds to a user-defined population with potentially different characteristics.

The stability of the weighting procedure is reflected in the effective sample size (ESS). For target populations close to the observed mixture, ESS remains relatively high and estimates are stable. As the target population diverges from the observed data, weights become more concentrated and ESS decreases, leading to increased variability. These findings emphasize that meaningful causal interpretation requires careful alignment between the target population and the support of the available data.

6.4 Bayesian Nonparametric Uncertainty

To quantify uncertainty in the weighted estimates, we employ a design-based Bayesian nonparametric approach based on Dirichlet weighting. This approach induces a posterior distribution over the treatment effect by placing a prior on the empirical weights assigned to reconstructed individuals.

For the KM-GPT pipeline, the posterior distribution of $\Delta(t_0)$ is concentrated and lies entirely above zero, indicating strong evidence for a positive treatment effect under the target population. In contrast, the RESOLVE-IPD posterior is wider and includes zero, reflecting greater uncertainty and weaker evidence for a treatment benefit.

These differences illustrate that uncertainty is not only driven by sample size but also by the structure of the reconstructed data and the weighting scheme. The design-based Bayesian nonparametric approach captures variability arising from the finite sample and the reweighting process, providing a more complete characterization of uncertainty than point estimates alone.

Importantly, this approach treats observed survival times as fixed and focuses on uncertainty in the population representation. As a result, it complements, rather than replaces, model-based approaches that explicitly account for uncertainty in the survival distribution.

6.5 Model-Based vs Design-Based BNP (NTR)

We compare the design-based Bayesian nonparametric approach with a model-based alternative based on a neutral-to-the-right (NTR) hazard formulation, applied to the KM-GPT reconstructed data.

The NTR model produces systematically moderated estimates of the treatment effect relative to the design-based approach. The posterior mean for $\Delta(t_0)$ is lower than the weighted Kaplan–Meier estimate but remains within the range of design-based Bayesian nonparametric uncertainty.

This difference reflects a fundamental distinction between the two approaches. The design-based method captures uncertainty through reweighting of observed data, while treating survival outcomes as fixed. In contrast, the NTR model directly represents the survival process through a stochastic hazard function, introducing smoothing that reduces sensitivity to reconstruction noise and extreme observations.

As a result, differences between design-based and model-based estimates are not merely statistical, but reflect alternative representations of the survival process.

t_0	$S_0(t_0)$	$S_1(t_0)$	$\Delta(t_0)$	SD	2.5%	97.5%
12.0	0.285	0.441	0.157	0.033	0.089	0.220

Table 2: Posterior summary of the treatment effect under the NTR hazard-based Bayesian nonparametric model.

Overall, this comparison demonstrates that Bayesian nonparametric inference is not unique: different formulations—design-based versus model-based—can yield materially different conclusions even when applied to the same reconstructed data. This highlights the im-

portance of considering multiple inferential frameworks when assessing robustness in causal survival analysis.

7 Discussion and Conclusions

7.1 Main Findings

This work develops a unified pipeline for causal survival inference using reconstructed individual patient data (IPD), integrating covariate augmentation, target-population reweighting, and Bayesian nonparametric inference. The results highlight several key insights.

First, reconstruction method has a measurable impact on downstream causal estimates. Even when applied to the same underlying studies, KM-GPT and RESOLVE-IPD produce different reconstructed datasets, leading to differences in both point estimates and uncertainty. This demonstrates that reconstruction should not be treated as a neutral preprocessing step, but rather as a source of variability that propagates into inference.

Second, target-population weighting plays a central role in defining the causal estimand. Naive pooled estimates reflect a mixture of study populations and are therefore difficult to interpret. By contrast, reweighting allows treatment effects to be evaluated with respect to a specified population, but can substantially change both the magnitude and direction of the estimated effect. This underscores that causal conclusions are inherently tied to the choice of target population.

Third, Bayesian nonparametric uncertainty quantification provides a more complete picture of inference. The design-based approach captures variability arising from finite samples and reweighting, revealing important differences in statistical confidence across reconstruction pipelines. In particular, the KM-GPT-based analysis suggests a consistently positive treatment effect, while the RESOLVE-IPD-based analysis exhibits greater uncertainty and weaker evidence.

Finally, the comparison between design-based and model-based Bayesian nonparametric methods shows that modeling assumptions about the survival process can materially affect conclusions. The NTR model produces systematically moderated treatment effect estimates relative to the design-based approach, indicating that uncertainty in the survival model itself is an important component of inference.

Taken together, these findings demonstrate that causal survival analysis using reconstructed data is a multi-stage process in which reconstruction, population definition, and modeling choices all contribute to the final estimate.

7.2 Limitations

Several limitations of the proposed framework should be noted.

First, individual-level covariates are not directly observed and must be generated from trial-level summary statistics. While the augmentation procedure accurately reproduces marginal distributions, it does not capture the joint dependence structure between covariates, which may affect the validity of reweighting in more complex settings.

Second, the target population is defined through a limited set of summary statistics. This may not fully characterize the underlying population of interest, and different choices of target summaries can lead to different causal estimates. In addition, extreme target populations can lead to unstable weights and reduced effective sample size, limiting the reliability of the resulting estimates.

Third, reconstructed IPD are inherently approximate. Differences between KM-GPT and RESOLVE-IPD highlight that reconstruction uncertainty can propagate into downstream inference, and the true underlying data remain unobserved.

Fourth, the design-based Bayesian nonparametric approach captures uncertainty in the weighting scheme but does not model the survival process itself. Conversely, the NTR model introduces additional modeling assumptions through the specification of the hazard process and prior structure. As a result, differences between design-based and model-based estimates reflect both statistical uncertainty and sensitivity to modeling assumptions.

Overall, these limitations emphasize that causal inference from reconstructed survival data requires careful consideration of both data quality and modeling choices, and that results should be interpreted in light of these uncertainties.

7.3 Practical Implications

The proposed framework provides a practical approach for conducting causal survival analysis in settings where only reconstructed individual patient data (IPD) are available. By integrating covariate augmentation, target-population reweighting, and Bayesian nonparametric inference, the pipeline enables analysts to move beyond descriptive survival comparisons and obtain interpretable causal estimates.

A key implication is that analysts should explicitly define the target population of interest. Naive pooling of reconstructed data can lead to misleading conclusions, as the resulting estimate reflects an implicit mixture of study populations. Reweighting allows treatment effects to be aligned with clinically relevant populations, but requires careful diagnostic checks, such as monitoring effective sample size (ESS), to ensure stability.

In addition, the results highlight the importance of evaluating sensitivity to reconstruction methods. Applying multiple reconstruction pipelines, when available, provides a simple but effective way to assess robustness of downstream conclusions. Differences across pipelines should be treated as a form of uncertainty rather than ignored.

Finally, the comparison between design-based and model-based Bayesian nonparametric approaches suggests that multiple inferential frameworks may be necessary to fully characterize uncertainty. In practice, reporting results from both approaches can provide a more comprehensive assessment of the treatment effect and its robustness to modeling assumptions.

7.4 Future Work

Several directions for future research follow naturally from this work.

First, more realistic modeling of covariate structure could improve the validity of the augmentation step. Incorporating dependence between covariates, for example through copula-

based methods or hierarchical models, would allow better approximation of the true joint distribution.

Second, joint modeling of reconstruction and inference represents an important extension. Rather than treating reconstructed IPD as fixed input, future approaches could propagate uncertainty from the reconstruction stage directly into downstream causal estimates.

Third, extending the framework beyond fixed time horizons would provide a more complete characterization of treatment effects. Modeling full survival curves or functionals such as restricted mean survival time (RMST) would allow richer comparisons between treatment groups.

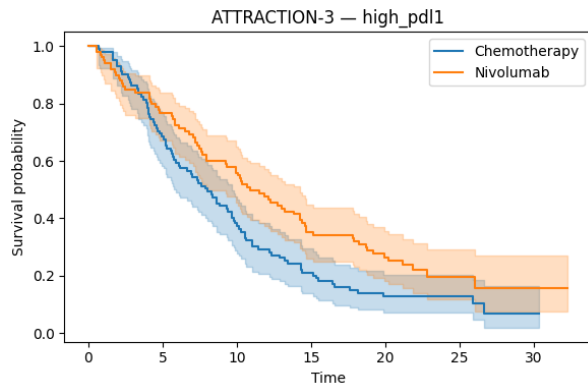
Fourth, further development of model-based Bayesian nonparametric approaches could improve flexibility and interpretability. Extensions of NTR-based or other hazard-based Bayesian nonparametric survival models that incorporate covariates or allow for hierarchical structure across studies may better capture heterogeneity in multi-trial settings.

Finally, applying the framework to a broader range of clinical areas and larger collections of studies would help assess its scalability and generalizability. As reconstruction methods continue to improve, combining them with robust downstream inference techniques offers a promising direction for expanding the scope of evidence synthesis in settings with limited data availability.

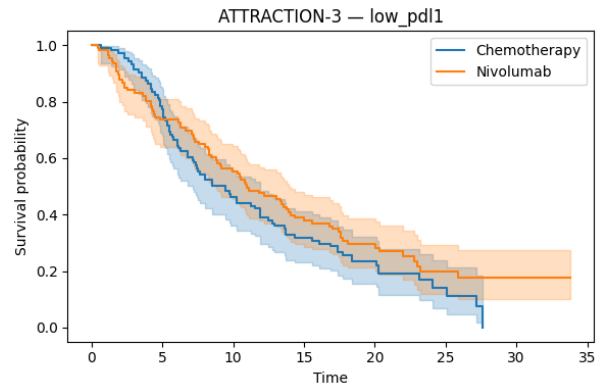
References

- [1] Yao Zhao, Haoyue Sun, Yantian Ding, and Yanxun Xu. Km-gpt: An automated pipeline for reconstructing individual patient data from kaplan–meier plots. *arXiv preprint arXiv:2509.18141*, 2025. URL <https://arxiv.org/abs/2509.18141>.
- [2] Lang Lang et al. Resolve-ipd: A high-fidelity approach to reconstructing individual patient data from kaplan-meier curves with subgroup information. *arXiv preprint arXiv:2511.01785*, 2024. URL <https://arxiv.org/pdf/2511.01785>.
- [3] K. Kato, B. C. Cho, M. Takahashi, M. Okada, C. Y. Lin, K. Chin, et al. Nivolumab versus chemotherapy in patients with advanced oesophageal squamous cell carcinoma refractory or intolerant to previous chemotherapy (attraction-3): A multicentre, randomised, open-label, phase 3 trial. *The Lancet Oncology*, 20(11):1506–1517, 2019. URL <https://pubmed.ncbi.nlm.nih.gov/31582355/>.
- [4] J. Huang, J. Xu, Y. Chen, W. Zhuang, Y. Zhang, Z. Chen, et al. Camrelizumab versus investigator’s choice of chemotherapy as second-line therapy for advanced or metastatic oesophageal squamous cell carcinoma (escort): A multicentre, randomised, open-label, phase 3 study. *The Lancet Oncology*, 21(6):832–842, 2020. URL <https://pubmed.ncbi.nlm.nih.gov/32416073/>.
- [5] Toshihiko Doi, Jaafar Bennouna, Lin Shen, Peter C. Enzinger, Ruixue Wang, Ildiko Csiki, et al. Keynote-181: Phase 3, open-label study of second-line pembrolizumab vs single-agent chemotherapy in patients with advanced/metastatic esophageal adenocarcinoma. *Journal of Clinical Oncology*, 34(15_suppl):TPS4140, 2016. URL https://ascopubs.org/doi/10.1200/JCO.2016.34.15_suppl.TPS4140.
- [6] K. W. Rott et al. Causally interpretable meta-analysis combining aggregate and individual participant data. *American Journal of Epidemiology*, 194(7):2060–2068, 2025. URL <https://academic.oup.com/aje/article/194/7/2060/7762598>.
- [7] S. Ying et al. Summary-statistics-based causal inference under covariate shift: Methods and applications. *arXiv preprint arXiv:2603.02474*, 2025. URL <https://arxiv.org/pdf/2603.02474v1>.

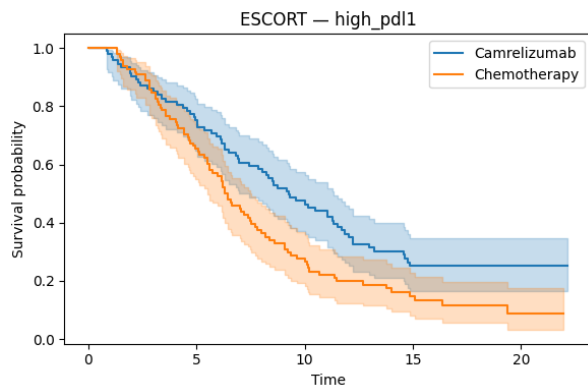
Appendix: Figures



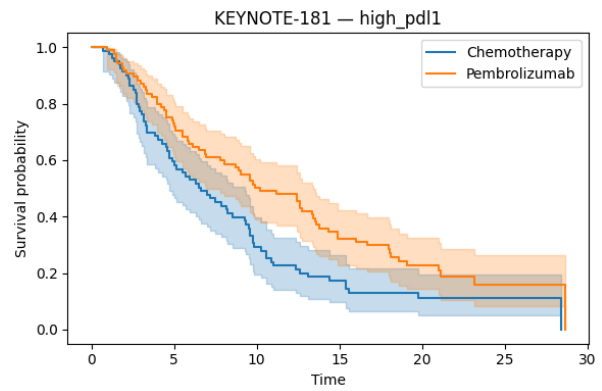
(a) ATTRACTION-3 (high PD-L1)



(b) ATTRACTION-3 (low PD-L1)



(c) ESCORT (high PD-L1)



(d) KEYNOTE-181 (high PD-L1)

Figure A1: Kaplan–Meier survival curves across the reconstructed oncology trial datasets used in the real-data application.

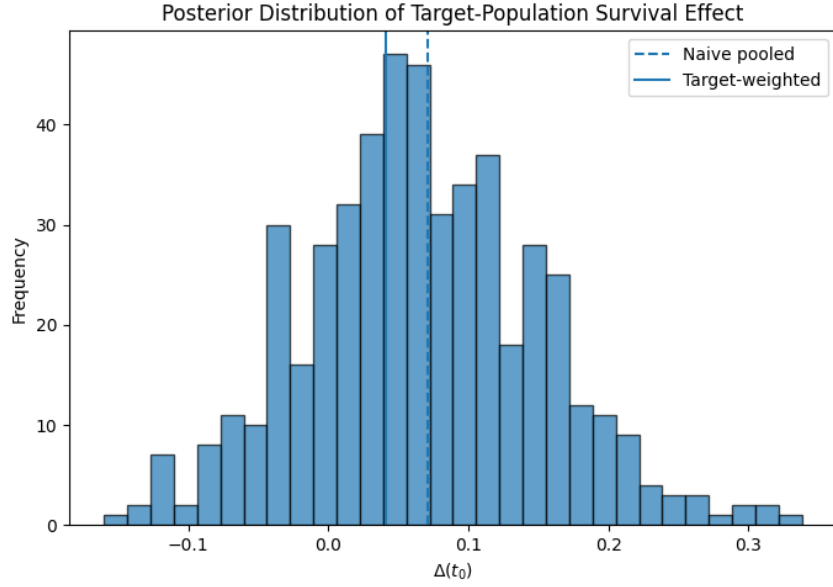


Figure A2: Posterior distribution of the treatment effect $\Delta(t_0)$ in the simulation-based Bayesian nonparametric analysis. Vertical lines indicate the naive pooled estimate and the target-weighted estimate.

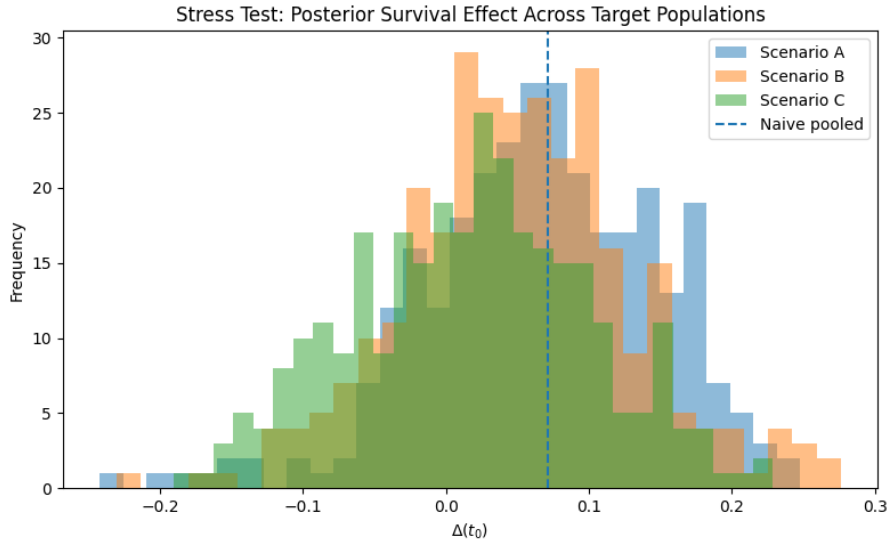


Figure A3: Posterior distributions of $\Delta(t_0)$ across stress-test target populations. Increasing covariate shift leads to broader uncertainty and less stable treatment effect estimates.

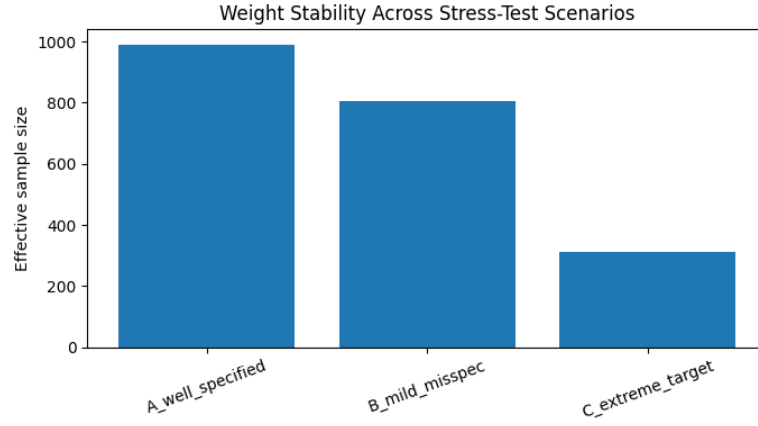


Figure A4: Effective sample size (ESS) across stress-test scenarios. ESS declines as the target population moves farther from the observed data distribution.

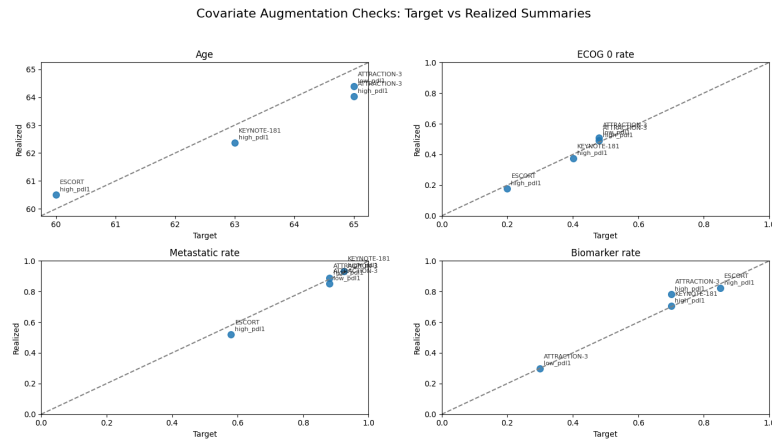


Figure A5: Validation of covariate augmentation by comparing target and realized covariate summaries. Points close to the identity line indicate accurate matching of reported auxiliary summaries.

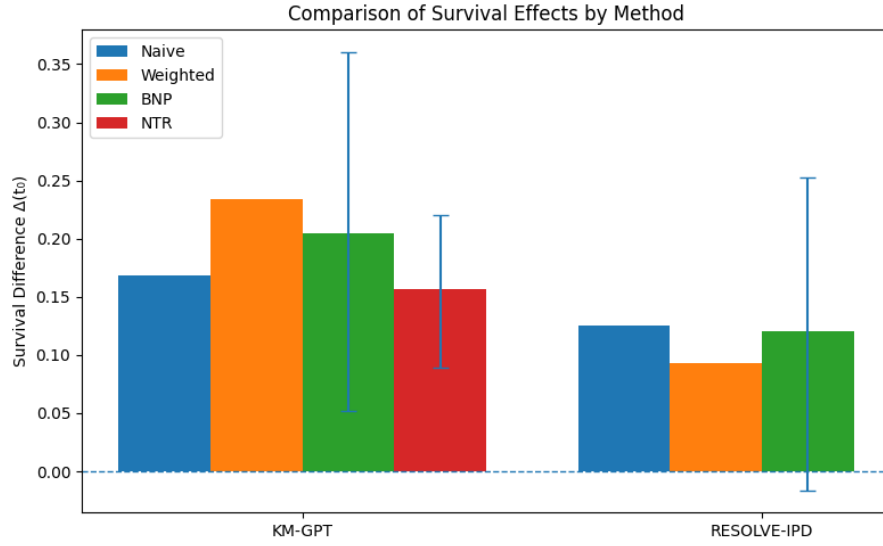


Figure A6: Comparison of treatment effect estimates across reconstruction pipelines and inference approaches. KM-GPT and RESOLVE-IPD differ in naive, weighted, and BNP-based estimates. The NTR model produces moderated treatment effect estimates due to smoothing of the survival process through hazard-based modeling.

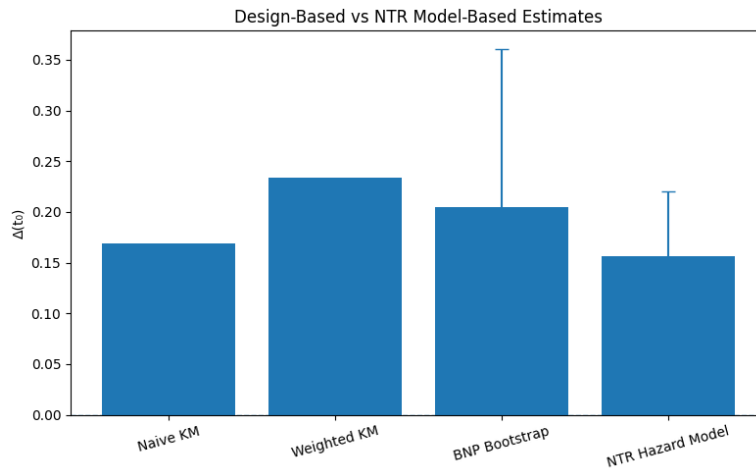


Figure A7: Comparison of design-based and model-based Bayesian nonparametric estimates for the KM-GPT pipeline. The NTR model produces moderated treatment effect estimates due to smoothing of the survival process through hazard-based modeling.